

Exp no: 7

Unification and Resolution – Exercises with Solutions

Exercise 1: Sentence Matcher

% Facts

sentence(the_cat_sits).

sentence(the_dog_runs).

% Query

% ?- sentence(X).

Output:

X = the_cat_sits ; X = the_dog_runs ; false.

Exercise 2: Compound Term Unification

% Facts

likes(john, food(pizza, spicy)).

likes(mary, food(pizza, sweet)).

% Query

% ?- likes(john, food(pizza, T)).

Output:

T = spicy.

Exercise 3: Conflict Resolution

% Facts

rule1(climate_change).

rule2(climate_change).

% Rule

conflict(X) :- rule1(X), rule2(X).

% Query

% ?- conflict(climate_change).

Output:

true.

```
% Query
SWI-Prolog console
File Settings Tools Debug GUI Help
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

12 ?- consult(['C:/Users/Lenovo/Documents/er.pl']).
true.

13 ?- likes(john, food(pizza, T)).
T = spicy.

14 ?- consult(['C:/Users/Lenovo/Documents/ep.pl']).
true.

15 ?- sentence(X).
X = the_cat_sits .

16 ?- onconsult(['C:/Users/Lenovo/Documents/eq.pl']).
Correct to: "consult(['C:/Users/Lenovo/Documents/eq.pl'])"?
Please answer 'y' or 'n'? yes
true.

17 ?- conflict(climate_change).
true.

18 ?- ◆
```

Result:

Thus the programs in prolog has been executed and the output is verified successfully.